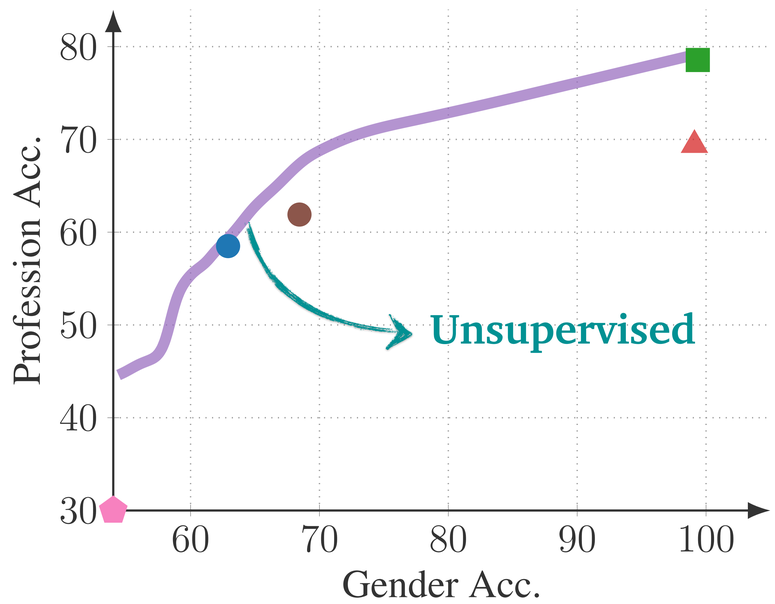
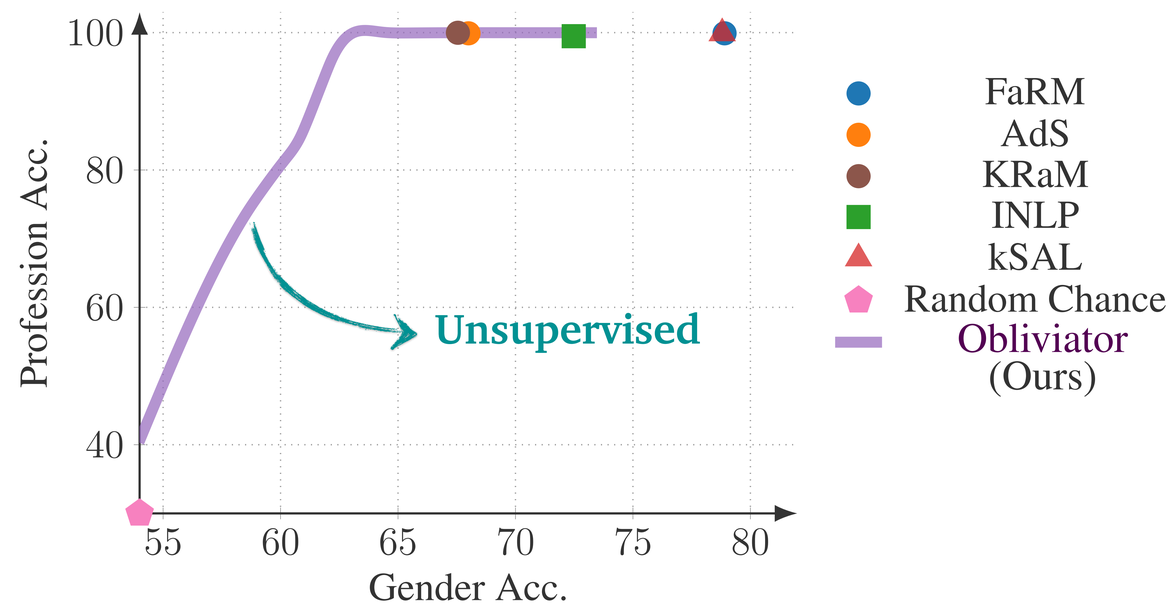
Obliviator Reveals : Effect of Target Task Labels on Erasure

Unsupervised

$$\inf_{\theta} \quad \mathrm{HSIC}(Z_\theta, S) - \left[\mathrm{HSIC}(Z^i_\theta, Y) + \mathrm{HSIC}(Z^i_\theta, X) + \mathrm{HSIC}(Z^i_\theta, X^i)\right]$$



BIAS IN BIOS



BIAS IN BIOS

Representation: Frozen

Representation: Finetuned

PLM : BERT

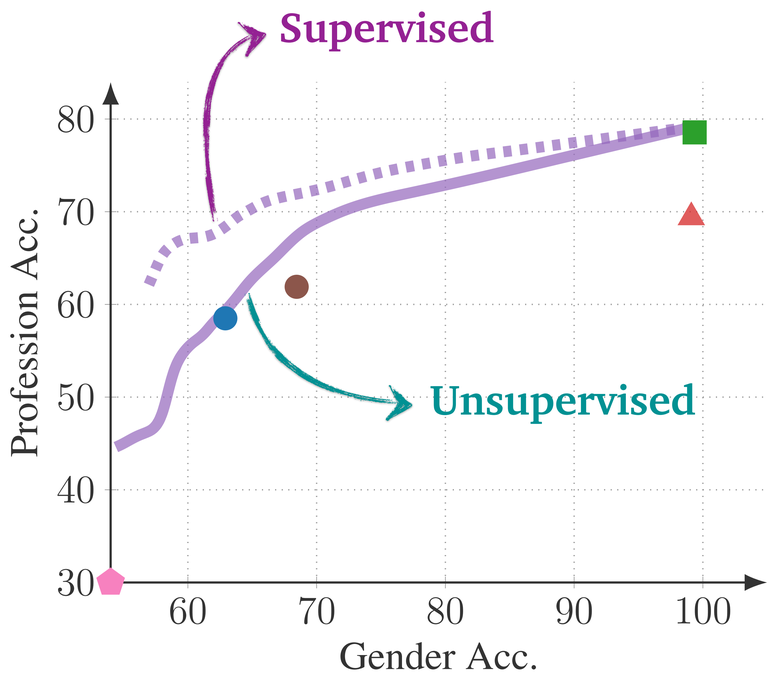
Utility : Profession

Unwanted : Gender

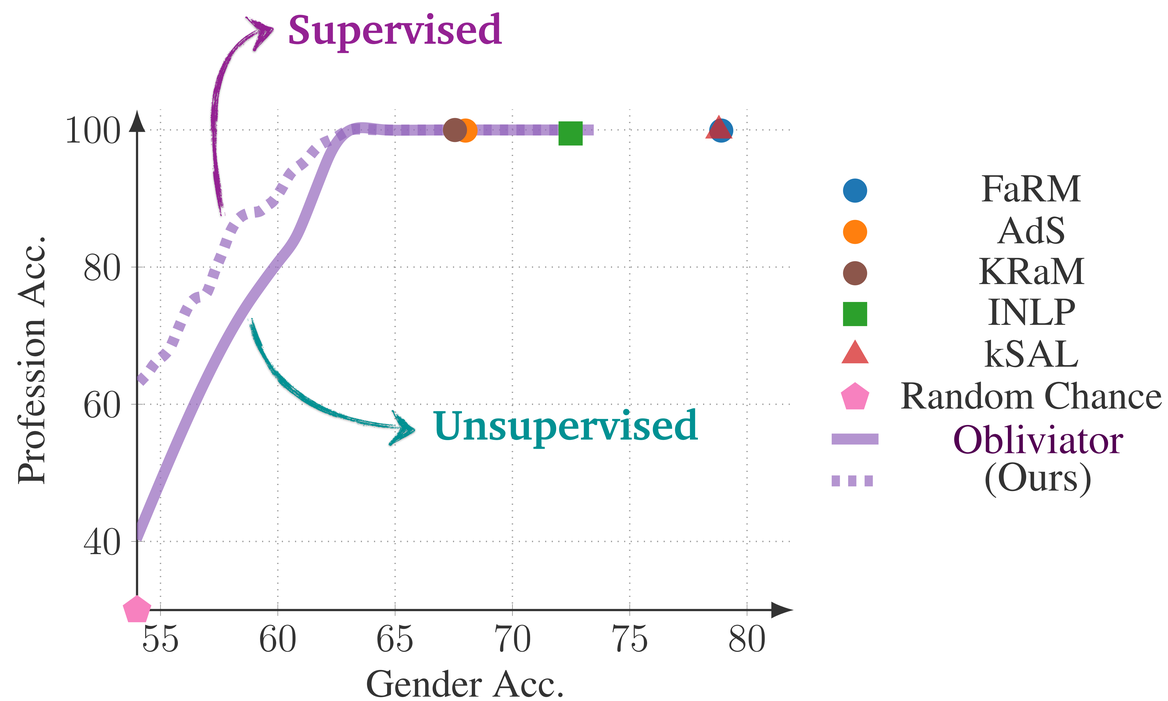
Obliviator Reveals : Effect of Target Task Labels on Erasure

Supervised

$$\inf_{\theta} \quad \text{HSIC}(Z_\theta, S) - \left[\text{HSIC}(Z_\theta^i, Y) + \text{HSIC}(Z_\theta^i, X) + \text{HSIC}(Z_\theta^i, X^i)\right]$$



BIAS IN BIOS



BIAS IN BIOS

Representation: Frozen

Representation: Finetuned

PLM : BERT

Utility : Profession

Unwanted : Gender